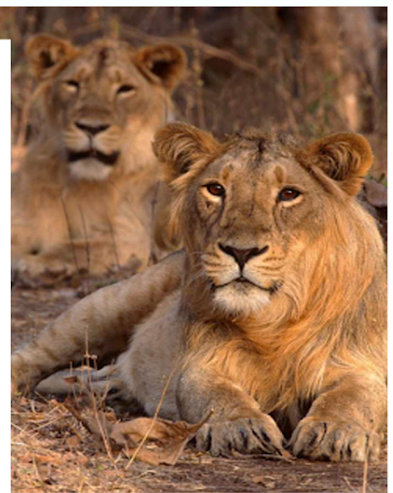
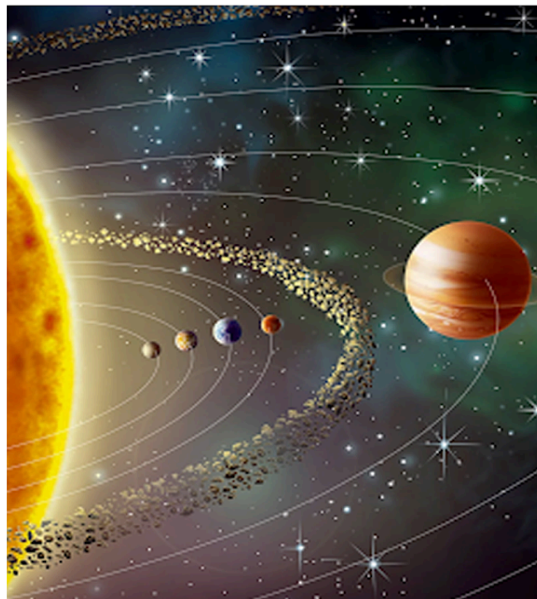


General Knowledge

World Geography

STUDY NOTES

GEOMORPHOLOGY: LANDFORMS(ARID)

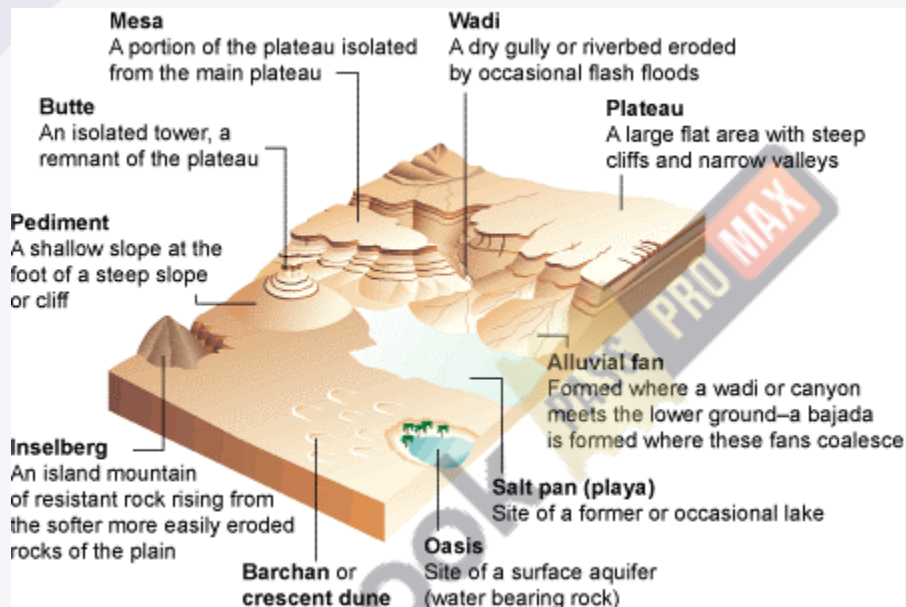


GEOMORPHOLOGY: LANDFORMS(ARID)

Arid Landforms

- Arid regions are regions with scanty rainfall. Deserts and Semi-arid regions fall under arid landforms.

Erosional Arid Landforms



Bolsons

- The intermontane basins in dry regions are generally known as bolsons.

Playas

- Three unique landforms viz. pediments, bajadas and playas are typically found in bolsons.
- Small streams flow into bolsons, where water is accumulated. These **temporary lakes are called playas**.

Pediments

- In form and function, there is no difference between a pediment and an alluvial fan; however, a pediment is an erosional landform while a fan is a constructional one.
- A true pediment is a rock-cut surface at the foot of the mountain.

Bajada

- Bajadas are **moderately sloping depositional plains** located between pediments and playa.
- Several alluvial fans coalesce to form a bajada.

Deflation basins

- Deflation basins, called blowouts, are hollows formed by the removal of particles by wind. Blowouts are generally small but may be up to several kilometres in diameter.

Mushroom rocks

- A mushroom rock also called a **rock pedestal or a pedestal rock**, is a naturally occurring rock whose shape, as its name implies, resembles a mushroom.
- The rocks are deformed in many different ways: by erosion and weathering, glacial action, or from a sudden disturbance. Mushroom rocks are related to, but different from, yardang.

Inselbergs

- A **monadnock or inselberg** is an isolated hill, knob, ridge, outcrop, or small mountain that rises abruptly from a gently sloping or virtually level surrounding plain.

Demoiselles

- These are rock pillars which stand as resistant rocks above soft rocks as a result of differential erosion of hard and soft rocks.

Zeugen

- A table-shaped area of rock found in arid and semi-arid areas formed when the more resistant rock is reduced at a slower rate than softer rocks around it.

Yardangs

- The ridge of rock is formed by the action of the wind, usually parallel to the prevailing wind direction.

Wind bridges and windows

- Powerful wind continuously abrades stone lattices, creating holes. Sometimes the holes are gradually widened to reach the other end of the rocks to create the effect of a window—thus forming a wind window. Window bridges are formed when the holes are further widened to form an arch-like feature.

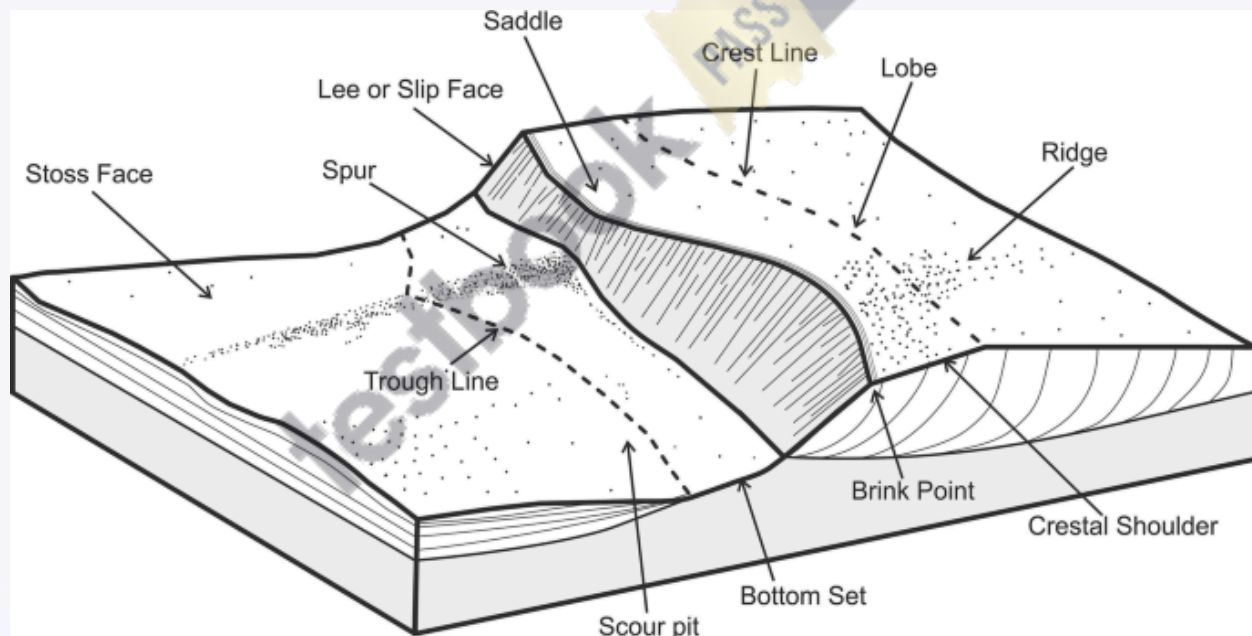


Arid Depositional Landforms

- The depositional force of wind also creates landforms. These are as follows.

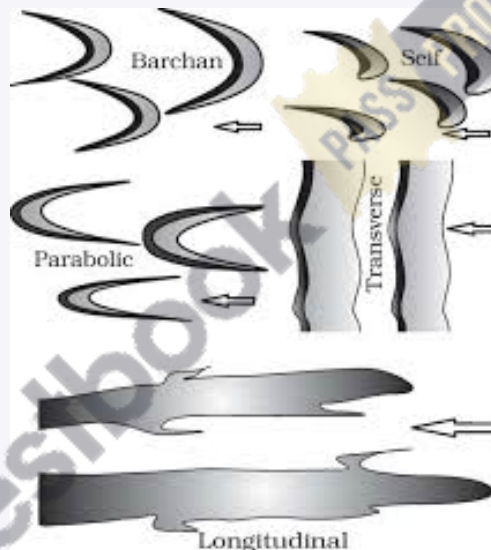
Ripple Marks

- These are depositional features on a small scale formed by saltation (the transport of hard particles over an uneven surface in a turbulent flow of air or water).

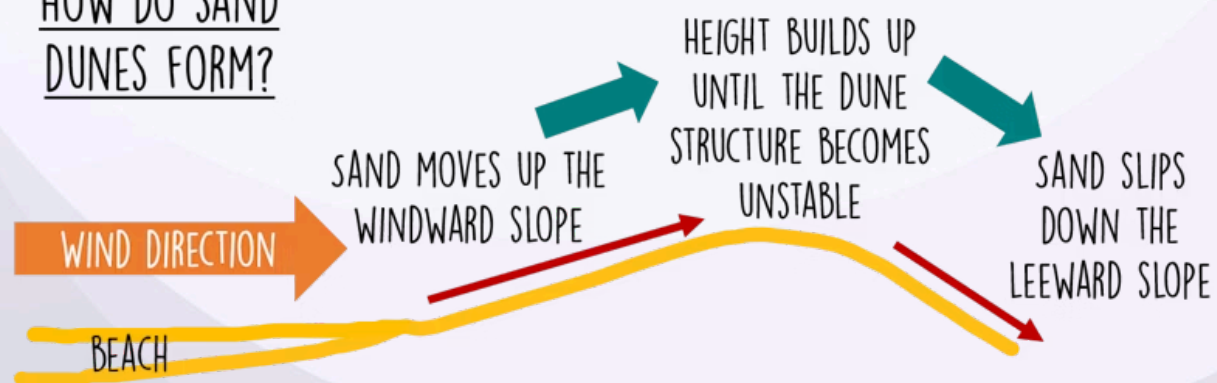


Sand Dunes

- Sand dunes are heaps or mounds of sand found in deserts. Generally, their heights vary from a few metres to 20 metres, but in some cases, dunes are several hundred metres high and 5 to 6 km long.
 - **Barchan Dunes:** Crescent-shaped dunes with tips pointing downwind.
 - **Transverse Dunes:** Long, straight ridges of sand perpendicular to the wind direction.
 - **Seif Dunes:** Long, linear dunes that form in parallel rows, typically in desert regions with dominant wind directions.
 - **Star Dunes:** Dunes that radiate from a central point, formed in areas with winds coming from multiple directions.
 - **Parabolic Dunes:** U-shaped dunes with tips pointing upwind, often associated with vegetation.



HOW DO SAND DUNES FORM?



Loess

- In some parts of the world, windblown dust and silt blanket the land. This layer of fine, **mineral-rich** material is called loess.
- Extensive loess deposits are found in **northern China, the Great Plains of North America, central Europe, and parts of Russia and Kazakhstan.**
- The thickest loess deposits are near the **Missouri River** in the U.S. state of Iowa and along the **Yellow River in China.**
- **Loess accumulates, or builds up, at the edges of deserts.** For example, as the wind blows across the Gobi, a desert in Asia, it picks up and carries fine particles. These particles include sand crystals made of quartz or mica. It may also contain organic material, such as the dusty remains of skeletons from desert animals.
- Loess often develops into **extremely fertile agricultural soil.** It is full of minerals and drains water very well. It is easily tilled, or broken up, for planting seeds.
- **Loess usually erodes very slowly** – Chinese farmers have been working the loess around the Yellow River for more than a thousand years.